CCC-C-428a
OCTOBER 27, 1959
SUPERSEDING
Int. Fed. Spec. CCC–C–00428 (ARMY–QMC)
July 9, 1959 and
Fed. Spec. CCC–D–746
February 17, 1939
(See 6.9)

FEDERAL SPECIFICATION

CLOTH, COTTON, DUCK; FIRE, WATER, WEATHER, AND MILDEW RESISTANT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers the requirements of Cloth, Cotton, Duck; Fire, Water, Weather, and Mildew Resistant Treated.

1.2 Classification.

1.2.1 *Types and classes.* — The finished fabric shall be of the following types and classes as specified (see 6.2) :

Type I. —Fabric finished in olive drab No. 7 color.
Type II.—Fabric finished in No. 20 deck grey color.
Class 1. —Regular finish.
Class 2. —Dry finish.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

CCC–C–419—Cloth, Cotton, Duck, Unbleached, Plied-Yarns (Army and Numbered).
CCC–D–761—Duck, Cotton, Plied-Filling-Yarns and Single Yarns (Flat Duck).
CCC–T–191—Textile Test Methods.
PPP–P–51 — Packaging, Packing, and Marking of Textile Fabrics.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–D–2384—Duck, Cotton, Plied and Single Yarns (High Sley).
MIL–D–3396—Duck, Cotton, Single and Plied Filling Yarns (Flat).

FSC 8305

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Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following publication of the issue in effect on date of invitation for bids, forms a part of this specification:

Munsell Book of Color.

(Copies may be obtained from the Munsell Color Company, 10 East Franklin Street, Baltimore, Md.)

3. REQUIREMENTS

3.1 Standard sample. — Unless otherwise specified, the cloth shall be equal to the standard sample in all respects not specifically covered by this specification (see 6.4).

3.2 Laboratory report approval and bid sample approval.—Unless otherwise specified at the time of submission of a bid, the prospective contractor shall submit to the contracting officer for approval, a sample 5 by 9 inches (minimum) and a certified copy of a recent laboratory report covering the type (i.e. flat duck, Army duck, numbered duck, etc.) which he proposes to deliver (see 6.2). Unless otherwise specified by the contracting officer, the bidder shall certify that the sample and laboratory report submitted cover duck which was finished in a mill where the finishing will be performed if the contract is awarded. The laboratory report shall contain test data which demonstrate that the finished fabric, which the contractor proposes to deliver, has been tested and found to comply with all requirements of this specification. Any of the following types of reports will be satisfactory from the standpoint of this requirement.

(*a*) An independent or commercial laboratory report.

(*b*) The prospective bidder's own laboratory report.

(*c*) A Government laboratory report from a contract within not more than 6 months of date of submission of bid.

The purpose of the above requirement is to assist the Government to determine the capability of bidders to manufacture a cloth meeting all the requirements of the specification. The submission of an acceptable sample under this requirement shall not be construed as relieving a contractor from subsequently meeting all requirements of the specification on all deliveries.

3.3 Formula approval. — The prospective contractor or his finishing plant shall submit to the contracting officer for approval, a list of materials which the prospective contractor or his finishing plant intends to use, stipulating grades and the standards of all materials including solvents. This list is for the information of the Government only and is not to be disseminated to the public (see 6.7). The use of compounds containing mercury in any form is prohibited.

3.3.1 *Disapproval of formula.* — In the event that the list of proposed materials is found to contain products which past experience has indicated may detract from the quality of the performance or which may present a toxic hazard to canvas goods workers, the proposed formulation shall be rejected (see 6.7).

3.3.2 *Approval of formula.*—Approval of the qualitative formula submitted under 3.3 shall in no way absolve the contractor from complying with the performance requirements of this specification.

3.4 Materials.

3.4.1 *Basic fabric.*—The fabric shall be as specified by the contracting agency (see 6.2).

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Sized fabric prior to finishing shall contain not more than 2.5 percent starch and protein materials and not more than 5.0 percent total size, finishing, and other nonfibrous materials including starch and protein when tested as specified in table VII.

3.4.2 *Grey fabric weight.*—Unless otherwise specified in subsequent tables and paragraphs where reference is made to grey fabric weight, the minimum acceptable grey fabric weight as given in the applicable grey fabric specification shall be used to determine the requirements under these paragraphs.

3.4.3 *Compound.*—The compound to provide fire, water, weather, and mildew resistance shall be well ground and evenly blended. Compounds containing sulfur may be permitted when specifically indicated by name in formulas approved by the contracting officer. The compound shall be of the same ingredients as contained in the list of materials submitted and approved under 3.3. Compounds containing mercury in any form are prohibited.

3.5 Finished fabric.

3.5.1 *Construction.*—The number of yarns per inch of the finished cloth shall conform to the requirements of the specification covering the grey fabric specified except that a minus tolerance in the filling yarn will be as allowed as follows:

Designation	*Minus tolerance*
Flat ducks (requiring desizing	Up to 10 percent
Army duck	Up to 5 percent
No. 8 and lighter duck	Up to 5 percent
Heavier than No. 8 duck	Up to 3 percent

3.5.2 *Finished width.*—The width of finished cloth shall be as specified in the contract or order (see 6.2 and 6.8) with the following tolerances permitted:

Numbered duck

Width of duck, inches	*Tolerance, inches*
40 inches or less	– ½ inch + ¼ inch
Over 40 inches to 51 inches incl.	– ½ inch + ⅜ inch
Over 51 inches to 81 inches incl.	– ¾ inch + ½ inch
Over 81 inches	– ⅞ inch + ⅝ inch

Army duck

Width of duck, inches	*Tolerance, inches*
40 inches or less	– ¼ inch + ½ inch
Over 40 inches to 51 inches incl.	– ¼ inch + ⅝ inch
Over 51 inches to 81 inches incl.	– ⅜ inch + ¾ inch
Over 81 inches	– ½ inch + ⅞ inch

Flat and high sley ducks — up to 15 ounces
(Requiring desizing)

Width of duck, inches	*Tolerance, inches*
40 inches or less	– 1¾ inches + ⅜ inch
Over 40 inches to 51 inches incl.	– 2¼ inches + ½ inch
Over 51 inches to 81 inches incl.	– 2⅝ inches + ⅝ inch
Over 81 inches	– 2⅞ inches + ¾ inch

Flat and high sley ducks — over 15 ounces
(Requiring desizing)

Width of duck, inches	*Tolerance, inches*
40 inches or less	– 1½ inches + ⅜ inch
Over 40 inches to 51 inches incl.	– 1¾ inches + ½ inch
Over 51 inches to 81 inches incl.	– 2¾ inches + ⅝ inch
Over 81 inches	– 2⅝ inches + ¾ inch

3.5.3 *Breaking strength.* — The breaking strength of the finished fabric in both warp

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and filling directions shall be not less than that of the corresponding specification requirements for the grey fabric.

3.5.4 *Weight.*—The weight of the finished fabric shall be the actual weight of the grey fabric plus an allowance for the treatment in accordance with table I.

TABLE I.

Grey weight (ounces per sq. yd.)	Allowable maximum increase in weight due to treatment based on actual grey fabric weight (Percent)
12 and over	50
Under 12	65

3.5.5 *Color.*—The color shall be as specified for the applicable type and shall match the approved shade sample (see 6.2) when furnished by the contracting activity (see 6.4). The color shall be obtained with carbon black, or lamp black, inorganic or phthalocyanine pigments.

3.5.5.1 *Colorfastness.*—When a standard sample is available, the colored and finished samples shall show fastness to accelerated weathering equal to or better than the standard sample. When a standard sample is not available or is not referenced as a standard for colorfastness, the colored and finished fabrics shall show "good" colorfastness to accelerated weathering after 100 hours when tested in accordance with method 5671 as specified in table VII.

3.5.5.2 *Matching.*—The color of the untreated duck shall match the approved standard shade under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.5.6 *Crocking fastness.*

3.5.6.1 *For type I and type II fabrics.*

3.5.6.1.1 *Class 1.*—Fabrics with the regular finish shall show no more crocking on dry white cotton cloth than the Munsell neutral value of 5.0 when tested in accordance with table VII.

3.5.6.1.2 *Class 2.*—Fabrics finished with a dry finish for upholstery purposes shall show no more crocking on dry white cotton cloth than Munsell neutral value of 7.0 when tested in accordance with table VII.

3.5.7 *Flexibility.*—When tested in accordance with procedures specified in table VII, the finished fabric shall show a bending moment of not more than that specified in table II for the given untreated numbered or Army duck classification as applicable.

3.5.8 *Water resistance.*—The finished fabric shall have a maximum leakage of 50 ml. at the height indicated in table III when tested by the water permeability test specified in table VII. Where finishing according to this specification is to be contracted for by using Government furnished grey fabric, adjustment shall be made on the water permeability, such that the finished fabric may exceed the 50 ml. requirement but not to exceed 25 percent of the grey fabric permeability. The fabric shall conform to the same maximum leakage requirements after the accelerated weathering test specified in table VII.

3.5.8.1 *Flat ducks.*—The requirements for resistance to water permeability shall not be applicable to the flat ducks of Federal Specification CCC–D–761 and Military Specification MIL–D–3396 with the exception of the 14.90 and 17.55 oz. flat ducks of Military Specification MIL–D–3396, to which the requirements of 3.5.8 are applicable.

3.5.9 *Fire resistance.*—When tested as specified in table VII the average time of

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TABLE II.—*Flexibility*

Designation	Applicable specifications	Initial bending moment maximum	After heating at 200°–205°F. (bending moment) maximum	At 0° ± 5°F. (bending moment) maximum
Numbered				
2	CCC–C–419	0.16	0.40	0.400
4	CCC–C–419	0.12	0.300	0.300
6	CCC–C–419	0.06	0.180	0.180
8	CCC–C–419	0.05	0.125	0.125
10	CCC–C–419	0.03	0.075	0.075
12	CCC–C–419	0.02	0.050	0.050
Army ducks				
8.25	CCC–C–419	0.013	0.032	0.032
9.85	CCC–C–419	0.013	0.032	0.032
12.29	CCC–C–419	0.013	0.032	0.032

For all other fabrics the flexibility requirements shall be as specified (see 6.2).

TABLE III.—*Height for test of water resistance*

Untreated weight (Oz/Sq Yd)	Hydrostatic height	
	Inches	*Centimeters*
Over 20	20	50.80
14 to 20	14	35.56
Under 14	8	20.32

TABLE IV.—*Average length of char*

Untreated weight (Oz/Sq Yd)	Maximum average length of char (inches)
Under 10	4.5
10 to 20	3.5
Over 20	2.0

flaming of the specimens tested for each unit of product shall not exceed 2 seconds after the burner flame is withdrawn, and no specimen shall in any case exceed 4 seconds of flaming after the burner flame is withdrawn. The average length of char shall not exceed the requirements of table IV.

The fabric shall conform to the same fire resistance requirements after the 100 hour accelerated weathering period of table VII and also after being exposed to water leaching (method 5832) when tested as specified in table VII.

3.5.10 *Mildew resistance.*

3.5.10.1 *For type I fabrics.*—The fabric shall be made mildew resistant by proper deposition within the fabric of a fungicide which is currently approved for this use by the procuring agency. Unless otherwise specified, the fungicide shall be one of the following in the specified concentration:

a. Copper-8-quinolinolate to effect a deposit, a minimum of 1.0 percent, and a maximum of 1.5 percent.

b. Copper naphthanate combined with copper-8-quinolinolate, minimum 0.3 percent copper to maximum 1.3 percent as copper naphthanate, and a minimum of 0.28 percent copper-8-quinolinolate to a maximum of 1.0 percent copper-8-quinolinolate.

c. Combination of copper-hydroxy naphthanate and copper-8-quinolinolate, a

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minimum of 0.35 percent to a maximum of 1.3 percent copper as copper-hydroxy naphthenate and a minimum of 0.28 percent to a maximum 1.0 percent of copper-8-quinolinolate.

The fungicide shall be uniformly applied and the fabric shall be well penetrated by fungicide so that specimens leached in accordance with table VII (method 5832) shall show no more than traces of any organism when observed with a hand lens of approximately 3X magnification after being tested as specified in table VII. Total copper content and the copper-8-quinolinolate content shall be determined in accordance with table VII.

3.5.10.2 *For type II fabrics.*—The fabric shall be made mildew resistant by proper deposition within the fabric of a fungicide which is currently approved for this use by the procuring agency. Two specimens shall be evaluated. The first specimen shall be the treated cloth as received and the second specimen shall be the treated cloth after the accelerated weathering test followed by the leaching resistance test as specified in table VII. Neither specimen shall show more than traces of mildew when tested by the method specified in table VII. The treated cloth shall not lose more than 15 percent of its strength due to mildew attack of specimens tested as received and after accelerated weathering followed by the leaching resistance test (5832) when tested as specified in table VII.

3.5.11 *Resistance to accelerated weathering.*

3.5.11.1 *Army and numbered ducks.*—The breaking strength of the finished fabrics under 15 ozs/sq yd or under the specified untreated weight shall retain not less than 75 percent of their original breaking strength or the breaking strengths specified for the grey material in the base fabric specification after exposure to 100 hours accelerated weathering as specified in table VII.

3.5.11.2 *Alternate tentage ducks.* — The breaking strength requirements for the material listed herein shall be the same as specified in 3.5.11.1 for Army and numbered duck:

MIL–D–2384—Type I, Class 1, 10.75 ounce (High sley)
—Type I, Class 2, 14.75 ounce (High sley)
MIL–D–3396—Type I, 14.90 ounce (Flat duck)
—Type I, 17.55 ounce (Flat duck)

3.5.11.3 *Flat ducks.* — The breaking strength for flat ducks except those listed in 3.5.11.2 shall be not less than 80 percent of their original breaking strength or the breaking strengths specified for the grey material in the base fabric specification after exposure to the 200 hours accelerated weathering when tested in accordance with table VII.

3.6 **Workmanship.** — The finished fabric shall be non-hygroscopic, clean, uniform in color, free from objectionable odor and shall not show two-sidedness other than may be attributed to difference in weave. The occurrence of defects shall not exceed the levels set by the applicable acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Civil agencies.

4.1.1 *Sampling.*

4.1.1.1 *Lot.* — Unless otherwise specified, all material of the same type and class presented at one time shall be considered a lot for purposes of inspection and tests.

4.1.1.2 *Sampling for visual and dimensional inspection.*—Unless otherwise specified, a random sample of cloth shall be taken from each lot in accordance with table V for inspection for visual and dimensional characteristics. The unit of product for this examination shall be 1 linear yard.

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TABLE V.—*Sampling for visual and dimensional inspection*

Lot size (yards)	No. rolls in inspection sample	Sample size * (yards)	Accpt. No. major defects	Reject. No. major defects	Accpt. No. major & minor defects	Reject. No. major & minor defects
300 and under	2	35	0	1	3	4
301 to 500	2	50	1	2	4	5
501 to 800	3	75	1	2	6	7
801 to 1,300	4	110	2	3	8	9
1,301 to 3,200	5	150	3	4	11	12
3,201 to 8,000	7	225	4	5	17	18
8,001 to 22,000	10	300	5	6	20	21
22,001 to 110,000	13	450	7	8	29	30

* Sample size in yards divided by number of rolls in inspection sample equals number of yards to be inspected in each roll.

4.1.1.3 *Sampling for test.*—Unless otherwise specified, random samples of duck cloth shall be taken from each inspection lot in accordance with the following table:

TABLE VI.—*Sampling for tests*

Number of rolls in lot	Number of rolls in sample*	Acceptance number (defective)	Rejection No. (defective)
8 and under	2	0	1
9 to 15	3	0	1
16 to 40	4	0	1
41 and over	5	0	1

* An equivalent of 5 square yards shall be taken from each roll for laboratory tests.

4.1.2 *Inspection.*—Each of the samples selected in accordance with table V or as specified shall be visually and dimensionally inspected to verify conformance with the requirements of this specification.

4.1.3 *Tests.*—Each of the samples selected in accordance with table VI, or as specified, shall be subjected to the tests specified in table VII.

4.1.4 *Compliance criteria.*

4.1.4.1 *Inspection.*—Any piece in the sample for inspection failing to conform to the requirements for which inspection was made (see 4.1.2), shall be considered defective. If the number of defects exceeds the acceptance number for that sample specified in table V, the entire lot from which the sample was taken shall be considered as having failed to meet the requirements of this specification.

4.1.4.2 *Examination.*—Yard by yard examination shall be in accordance with paragraph 4.3.1.1 with the following exception: Delete sub-paragraphs "a. and b.". Examination for overall defects shall be in accordance with paragraph 4.3.1.2. Examination for length of individual pieces shall be in accordance with paragraph 4.3.1.3.1 and examination for total yardage in sample shall be in accordance with paragraph 4.3.1.3.2.

4.1.4.3 *Tests.*—Any unit of product in the sample for test failing to conform to the re-

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quirements for which tests were made (see 4.1.3), shall be considered defective. If the number of defective units of products in the sample for test exceeds the acceptance number for that sample specified in table VI, the entire lot of cloth from which the sample was taken shall be considered as having failed to meet the requirements of this specification.

4.2 Military agencies.

4.2.1 *Contractors' inspection responsibility.*—Unless otherwise specified, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and made available to the Government as specified in the contract or order.

4.2.2 *Preacceptance inspection.*

4.2.2.1 *Laboratory report approval.*—The data contained in the report submitted in accordance with 3.2 shall be obtained in accordance with test methods specified in table VII.

4.3 Inspection for acceptance.—Inspection for military procurements shall be in accordance with the provisions of Military Standard MIL–STD–105, except where otherwise indicated.

4.3.1 *Examination of the end item.*—The cloth shall be examined to determine conformance with the requirements of this specification. Defects found during this examination shall be classified in accordance with 4.3.1.1, 4.3.1.2 and 4.3.1.3.

4.3.1.1 *Yard by yard examination.*—The defects listed in tables VIA and VIB as applicable shall be counted regardless of their proximity to each other except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The unit of product for this examination shall be 1 linear yard. The acceptable quality levels for this examination shall be as follows:

a. Finished duck when the grey duck is furnished by the Government (table VIB).
   (1) Width of fabric 56 inches or less: 0.25 major defects and 1.5 total defects (major and minor combined) per 100 units (yards).
   (2) Width of fabric more than 56 inches: 0.40 major defects and 2.5 total defects (major and minor combined) per 100 units (yards).

b. Finished duck when grey duck is contractor furnished (table VIA and table VIB combined).
   (1) Width of fabric 56 inches or less: 0.40 major defects and 2.5 total defects (major and minor combined) per 100 units (yards).
   (2) Width of fabric more than 56 inches: 0.65 major defects and 4.0 total defects (major and minor combined) per 100 units (yards).

4.3.1.2 *Overall examination.* — Applicable when grey duck is supplied either by the Government or by the contractor. Each defect listed below shall be counted no more than once in each piece examined. The unit of product for the examination shall be one piece. The sample size and acceptance number shall be as shown in table VIC.

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TABLE VI A.—*Applicable to finished duck when grey goods are supplied by the contractor*

Glossary No. Fed. Std. 104	Defect	Classification	
		Major	Minor
55, 56, 90, 91, 92	Hole, cut, tear or break (including selvage)—Any	X	
78	Smash—Any	X	
79, 89	Abrasion mark, bruise, tender or weak spot—Any	X	
11, 33	Broken or missing yarns: Three or more contiguous regardless of length Two contiguous, 2 inches or more missing Two contiguous, less than 2 inches or single 2 inches or more missing	 X X	 X
9	Fine or light filling bar, light place—clearly visible*	X	
2	Course filling bar or heavy place—clearly visible* and exceeding 2 linear inches		X
5, 26, 27, 77	Slough off, jerked in filling, slubs, gouts, knots—more than three times the thickness of normal yarns		X
9, 12	Open place or crack—clearly visible*	X	
13, 34	Floats—multiple ½ inch or more in combined warp and filling directions, or harness skips Multiple less than ½ inch in combined warp and filling directions, or single 2 inches or more in length	X	 X
41	Reed mark or temple mark—resulting in a clear warpwise separation markedly wider than normal		X
40	Wrong draw—½ yard or more		X
	Width narrow or wide—beyond specified tolerances		X
49, 50	Slack or tight selvage resulting in pronounced slackness or waviness in cloth		X
51	Scalloped selvage—indentations exceeding ½ inch		X
48	Curled or folded selvage—½ yard or more		X
53, 54	Stringy, beaded, corded, loopy or rough selvages ½ yard or more, resulting in raised places along the selvage edge 2 or more times the normal fabric thickness, ½ yard or more		X
74, 75, 76	Spot, stain or streak 1 inch or more in combined directions—clearly visible* More than ¼ inch but less than 1 inch in combined directions clearly visible*	X	 X
55, 56, 89, 90, 91, 92	Hole, cut, tear or weak place (including selvage) caused in finishing operation. Clearly visible*	X	

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TABLE VI A.—*Applicable to finished duck when grey goods are supplied by the contractor* (Cont'd)

Glossary No. Fed. Std. 104	Defect	Classification	
		Major	Minor
48, 60	Selvage curled or folded, or crease-embedded. One half yard or more		X
74, 75, 76	Spots, stains, streaks, shade bar (caused in finishing process), 1 inch or more in combined directions clearly visible*	X	
	¼ inch but less than 1 inch in combined directions clearly visible*		X

* At normal inspection distance (approximately 3 feet).

TABLE VI B.—*Applicable to finished duck when grey duck is either Government or contractor furnished*

Examined	Defect	Classification	
		Major	Minor
Hole, cut, tear or weak place (including selvage) caused in finishing operation	Clearly visible*	X	
Selvage curled or folded or Crease-embedded	One half yard or more		X
Spots, stains, streaks, shade bar (caused in finishing process)	1 inch or more in combined directions clearly visible*	X	
	¼ inch but less than 1 inch in combined directions clearly visible*		X

* At normal inspection distance (approximately 3 feet).

Defects

Baggy, ridgy or wavy cloth.

Color not uniform, off shade (not within established tolerances), shaded end to end, side to side or side to center.

Poorly constructed—not firmly or tightly woven.

Overall application of compound not uniform.

Objectionable odor.

Overall uncleanliness.

Clearly noticeable crystallization of mildew inhibitor.

TABLE VI C.—*Sample size*

Lot size in yards	Sample size in pieces	Maximum No. of defects acceptable in sample
Up to 1,300[1]	3	0
1,301 to 3,200	5	0
3,201 to 8,000	7	0
8,001 to 22,000	10	0
22,001 to 110,000	15	0
110,001 and over	25	1

[1] If a lot contains fewer than 3 pieces each piece in the lot shall be examined.

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4.3.1.3 *Length examination.* — Applicable when grey duck is supplied either by the Government or by the contractor.

4.3.1.3.1 *Examination for length of individual pieces.*—Each piece in the sample shall be examined for gross length. Any gross length found to be less than the specified minimum length or more than maximum gross length permitted, or any gross length found to be more than 2 yards less than the gross length marked on the ticket, shall be considered a defect with respect to length. The unit of product for this examination shall be one piece. The sample size and acceptance number shall be as shown in table VI C.

4.3.1.3.2 *Examination for total yardage in sample.*—The lot shall be unacceptable if the total of the actual gross lengths of the pieces in the sample is less than the total of the gross lengths marked on the piece tickets. The pieces examined shall be those selected for examination of individual pieces.

4.3.2 *Inspection for preparation for delivery.*—The preservation, packaging, packing and marking shall be inspected to assure compliance with section 5.

4.4 Testing of the end item.—The physical and chemical values specified in section 3 (except where otherwise indicated) apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test methods. The method of testing specified in Federal Specification CCC–T–191, wherever applicable and as listed in table VII shall be followed. The tests shall be subdivided into two categories as follows:

a. Short term tests. — Tests to be performed on a lot by lot basis. These tests are designated by a single asterisk in table VII. The acceptable quality level for each characteristic shall be 6.5 test failures per 100 units. The sample size shall be in accordance with inspection level L–1, table III–B of Military Standard MIL–STD–105. The lot size shall be expressed in units of 1 linear yard each. The unit of product for testing purposes shall be 3 yards of finished duck and ½ yard of grey duck.

b. Long term tests. — Tests to be performed in accordance with table VII. These tests are designated by a double asterisk in table VII. The acceptable quality level for each characteristic shall be 6.5 test failures per 100 units and the sample size shall be in accordance with inspection level L–1, table III–B of Military Standard MIL–STD–105. Long term tests shall be performed on the basis of the following frequency:

(1) On the first lot.
(2) Once in every five lots thereafter.

The lot size shall be expressed in units of 1 linear yard each. The unit of product for testing purposes shall be 2 yards of finished duck and ½ yard of grey duck.

Failure to meet a requirement for a long term test will be cause for rejection of the lot in question and will result in the characteristic for which the failure occurred being placed on a lot-by-lot basis for test purposes until three consecutive lots shall have passed the requirement.

4.4.1 *Conditioning of samples.*—All samples shall be conditioned for at least 16 hours in an atmosphere having a relative humidity of 65 ± 2 percent and a temperature of 70° ± 2°F. before testing.

4.4.2 *Flexibility tests.*—The test shall be conducted in accordance with method 5202 with the following modifications:

Apparatus.—The bending moment required to deflect a 1-inch strip, ½-inch span, to a 20 degree deflection shall be measured. The rate of angular deflection shall be 60 degrees per minute.

Specimens.—Eight specimens 1 inch by 2 inches shall be cut, four with the long dimension in the warp direction and four in the filling direction, and pressed

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TABLE VII.—*Test methods*

* Short term test (see 4.4.a).
** Long term test (see 4.4.b).

Characteristics	Requirement reference [1]	Method number and paragraph reference [1]	Number of determinations per individual unit of product	Results reported as:
Weight*	3.4.2 3.5.4	5041	5	Average of 5 determinations to nearest whole number
Yarns per inch: *				
Warp	3.5.1	5050	5	Average of 5 determinations to nearest whole number for both warp and fill
Filling	3.5.1	5050	5	
Breaking strength: *				
Initial warp *	3.5.3	5100	5	Average of 5 determinations to nearest 1.0 lb. for warp and filling
filling	3.5.3	5100	5	
After accelerated weathering:**				
Warp	3.5.11.1 3.5.11.2 3.5.11.3	5100–5804 par. 4.4.3	5	Average of 5 determinations to nearest 1.0 lb. for both warp and filling
Filling	3.5.11.1 3.5.11.2 3.5.11.3	5100–5804 par. 4.4.3	5	
Flexibility:**				
Initial warp	3.5.7	5202, par. 4.4.2	4	Average of 4 determinations to nearest .001 lb. for both warp and filling
filling	3.5.7	5202, par. 4.4.2	4	
After heat aging:**				
Warp	3.5.7	5202, par. 4.4.2	4	Average of 4 determinations to nearest .001 lbs. for both warp and filling
Filling	3.5.7	5202, par. 4.4.2	4	
At low temperature:**				
Warp	3.5.7	5202, par. 4.4.2	4	Average of 4 determinations to nearest .001 lb. for both warp and filling
Filling	3.5.7	5202, par. 4.4.2	4	
Color*	3.5.5	par. 3.5.5.2	1	Satisfactory or unsatisfactory
Colorfastness to weathering**	3.5.5.1	5671	1	Standard sample rating of satisfactory or unsatisfactory. No standard sample: Excellent, good, fair, poor
Fastness to crocking*	3.5.6	5651	1	Satisfactory or unsatisfactory
Flame resistance:*				
Initial warp*	3.5.9	5903	4	Average of 4 determinations to nearest 0.1 inch char for both warp and filling
filling	3.5.9	5903	4	
After accelerated weathering:**				
Warp	3.5.9	5903, par. 4.4.3	4	Average of 4 determinations to nearest 0.1 inch char for both warp and filling
Filling	3.5.9	5903, par. 4.4.3	4	
After water leaching:[4] **				
Warp	3.5.9	5903, 5832	4	Average of 4 determinations to nearest 0.1 inch char for both warp and filling
Filling	3.5.9	5903, 5832	4	

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TABLE VII.—*Test methods* (Cont'd)

Characteristics	Requirement reference [1]	Method number and paragraph reference [1]	Number of determinations per individual unit of product	Results reported as:
Water resistance:[2] **				
Grey fabric	3.5.8	5516	3	Average of 3 determinations to nearest 1.0 cc
Treated fabric initially*	3.5.8	5516	3	
Treated fabric:**				
After accelerated weathering	3.5.8	5516–5804 par. 4.4.3	3	Average of 3 determinations to nearest 1.0 cc
Non-fibrous materials content [3]*	3.4.1	2611	2	Average of 2 determinations to nearest 0.1 percent
Copper anlysis*	3.5.10	2050 or 2051	2	Average to nearest .01 percent
Copper-8-quinolinolate*	3.5.10	par. 4.4.4	2	Average to nearest .01 percent
Mildew resistance for type I**	3.5.10.1	par. 4.4.5 5832	1	Growth or no growth
Mildew resistance for type II**				
1. As received:**				
A. Growth	3.5.10.2	par. 4.4.5	1	A. Growth or no growth
B. Loss in tensile strength.	3.5.10.2	par. 4.4.5	5	Average of 5 determinations to nearest percent
2. After accel. weathering and leaching:**				
A. Growth	3.5.10.2	par. 4.4.5 5832–5804	1	Growth or no growth
B. Loss in tensile strength.	3.5.10.2	par. 4.4.5 5832–5804	5	Average of 5 determinations nearest percent
Compound*	3.4.3 [5]	Std. comml.	1	Pass or fail

[1] Method numbers refer to Federal Specification CCC–T–191, paragraph and table references apply to this specification.

[2] Method 5516 shall be used except that the specimen shall be immersed in water at a temperature 75° to 85°F. to a depth of 2 inches for 2 hours and shall be tested while still wet. The hydrostatic head for testing shall be as specified in table III of this specification and time of exposure to this hydrostatic head shall be 10 minutes.

[3] The amount of nonfibrous materials shall be determined with the following formula:

$$\text{Total nonfibrous material, percent} = \frac{O-S+R}{O} \times 100$$

$$\text{Starch and protein material, percent} = \frac{O-S-W-C}{O} \times 100$$

[4] Leaching period shall be 24 hours.

[5] This requirement may be accepted on the basis of contractor's certification of compliance with specification requirements in lieu of test results.

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between two 6- by 6-inch glass plates weighted with a 20 pound weight for four hours at standard conditions.

4.4.2.1 *After heat aging.*—The specimen shall be tested for flexibility after being subjected to a temperature between 200° and 205° for 120 hours in a well-ventilated oven and then being allowed to cool at standard conditions for 16 hours.

4.4.2.2 *At low temperature.*—Flexibility test shall be conducted at a temperature of 0° ± 5°F. upon specimens which have been subjected to that temperature for not less than one hour.

4.4.3 *Accelerated weathering test.*

4.4.3.1 *Apparatus.*—Unless otherwise specified the apparatus shall be as specified in method 5804, with the following exceptions:

(*a*) Delete from title "(National Weathering Unit)."

(*b*) The filters shall be removed when testing the following materials:

CCC–C–419—All types Army, Numbered, and Tent-Duck

MIL–D–2384—Type I, class 1, 10.75 ounce (High sley)
—Type I, class 2, 14.75 ounce (High sley)

MIL–D–3396—Type I, 14.90 ounce (Flat duck)
—Type I, 17.55 ounce (Flat duck)

(*c*) Delete paragraph 3.1.1.5 in its entirety and substitute: "Water-spray nozzles shall be mounted vertically in the test chamber inside the specimen rack and so spaced that water shall strike the specimens evenly over their entire length in the form of a fine spray in sufficient volume to wet the specimens immediately on impact. The apparatus shall be so operated that the specimens are exposed to successive cycles of light without spray and light with spray. The width of the spray, at plane of impact, shall be not less than 6 inches nor more than 12 inches.

(*d*) Delete paragraphs 4.10 and 4.10.1 in their entirety and substitute: "The black panel thermometer shall be mounted on the test panel rack and readings taken at a point where maximum heat is developed and where water spray is not striking the panel, which temperature shall be 175°F. plus or minus 10° during any exposure cycle."

4.4.3.2 *Procedure.*—Two specimens, 9½ by 28 inches, shall be cut from the treated cloth under test. One specimen shall be cut with the long dimension in the direction of the warp; the other with the long dimension in the direction of the filling. In the event that the width of the treated cloth does not permit a full 28 inch specimen to be taken in the filling direction, two 9½- by 14-inch specimens shall be cut. The specimens shall be exposed one above the other in the quadrant of the accelerated weathering apparatus. The specimens shall be changed from the top to bottom racks approximately every 25 hours during the designated exposure period (see 3.5.11). At the conclusion of the exposure period, breaking strength and fire resistance specimens shall be cut from the exposed material. The four fire resistance specimens in each direction of the treated cloth shall be exposed to accelerated weathering in a vertical position. In conducting fire resistance tests on these specimens, two specimens shall be subjected to the test flame at the end which was at the top of the exposed material, and two shall be subjected to the test flame at the end which was at the bottom of the specimen. The lower edges of the fire resistance test specimens shall be trimmed if necessary so that a freshly cut end is exposed to the test flame.

4.4.3.3 The specimens shall be exposed for the designated period, after which they shall

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be allowed to dry, and then be conditioned at standard conditions before testing.

4.4.4 *Copper-8-quinolinolate analysis.* — The copper-8-quinolinolate content of the treated material shall be obtained using the following method:

4.4.4.1 *Reagents.*

10 percent sulfuric acid solution prepared by weight
Redistilled chloroform
25 percent ammonium hydroxide solution
Anhydrous sodium sulphate

4.4.4.2 *Apparatus.*

Balance
Potentiometric pH apparatus or pH indicator paper
250-ml. separatory funnel
Spectrophotometer capable of making measurements at 410 millimicrons
Beakers
100-ml. Erlenmeyer flasks

4.4.4.3 *Procedure.*—The sample shall be prepared as follows: Weigh to the nearest milligram a sample of approximately 1 g. cut into $\frac{1}{16}$ inch squares and immerse the weighed specimens in 25 ml. of 10 percent sulfuric acid contained in a 100-ml. Erlenmeyer flask. Heat the solution to 90° — 95°C. with shaking. Decant the acid extract into a 250-ml. beaker and repeat the extraction twice more with 15 ml. portions of the 10 percent sulfuric acid solution. Add to the combined acid extracts a sufficient quantity of 25 percent ammonium hydroxide solution to adjust the pH of the solution to the range of 5 to 7 as measured by either a potentiometric pH meter or pH indicator paper. Transfer the neutralized solution to a 250-ml. separatory funnel. Add 5 to 10 ml. of redistilled chloroform. Shake the separatory funnel vigorously for at least one minute. Allow the contents to settle for 1-2 minutes, or until the chloroform-water meniscus is clear. Collect the chloroform layer in a 100-ml. beaker. Several extractions are necessary for complete separation. Repeat the extractions twice, until the chloroform layer is colorless. Add approximately 10 g. of anhydrous sodium sulphate to the combined chloroform extracts. Stir thoroughly and allow to stand five minutes. Then carefully decant chloroform solution into a 100-ml. calibrated volumetric flask, washing the sodium sulphate thoroughly with several portions of chloroform. Dilute to mark with chloroform. Read the percent transmission of the above solution at 410 millimicrons on spectrophotometer, using as the solvent blank redistilled chloroform.

4.4.4.4 *Preparation of the standard curve.* —Dissolve 100 mg. of C.P. Copper-8-quinolinolate in 10 percent sulfuric acid solution. Pour into a 100-ml. volumetric flask and adjust the volume to the mark with the 10 percent sulfuric acid. Take aliquots of 1 ml. and whole multiples thereof for the points of the curve representing different concentrations of copper-8-quinolinolate. Pipette each aliquot into a 150-ml. beaker and dilute to 55 ml. with distilled water. Add 25 percent ammonium hydroxide solution until the pH of the solution is between 5 and 7 as measured by either a potentiometric pH meter or pH indicator paper. Proceed from this point as described in procedure above. The standard curve should cover a range of 1 mg. of copper-8-quinolinolate per 100 ml. of chloroform to 7 mg. of copper-8-quinolinolate per 100 ml. of chloroform.

4.4.4.5 *Calculation of results.*—Using the calibration curve, read the number of grams of copper-8-quinolinolate in the sample taken.

$$\text{Percent, Copper-8-Quinolinolate} = \frac{\text{gms. of Copper-8-Quinolinolate}}{\text{Weight of sample in gm.}} \times 100$$

$$\text{Percent, Cu as Cu-8-Quinolinolate} = \frac{\text{gms. Copper-8-Quinolinolate (18.06)}}{\text{Weight of sample in gm.}}$$

4.4.5 *Mildew resistance.*—The resistance to mildew shall be determined in accordance with the following procedure:

4.4.5.1 *Mildew resistance of textiles; mixed culture method.*

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4.4.5.1.1 *Test specimen.* — The specimen shall be as described in Method 5104 of Federal Specification CCC–T–191 cut with the long dimension parallel to the warp.

4.4.5.1.2 *Apparatus.*

Autoclave.—Autoclave which can be operated at a pressure of 15 pounds per square inch steam pressure and temperature of 121°C.

Petri dishes.—Petri dishes with an inside diameter of 10 centimeters.

Sterile room.—A dust-free sterile room or compartment in which to prepare inocula and make inoculations. Sterile lamp or antiseptic sprays may be used to maintain sterile conditions.

Incubator. — An incubation room or chamber maintained at a temperature of 28° to 30°C. and a relative humidity of 85 to 95 percent.

Visibility control.—Untreated material similar in all other respects to the treated material under test shall be inoculated and incubated with the test material. When untreated test material is unavailable, untreated material similar to the test material in composition, weight and form may be substituted. If this control material fails to lose 85 percent of the initial tensile strength or, for qualitative tests, does not show abundant mildew, the test shall be considered inconclusive and be repeated.

Culture media.—Test medium shall contain the ingredients in the preparations shown:

Potassium dihydrogen phosphate	–1.00g.
Ammonium nitrate	–3.00g.
Magnesium sulfate with 7 molecules of water of crystalization	–0.25g.
Potassium chloride	–0.25g.
Agar	–20.0g.
Distilled water	–1000ml.

Stock culture medium.—All stock cultures shall be maintained on Potato Dextrose Agar slants.

Organisms used shall be:

Chaetomium globosum. — Q.M.No. 459 (U.S.D.A. 1042.4)

Myrothecium verrucaria.—Q.M.No. 460

Trichoderma Sp — Q.M.No. 365 (ATCC 9645)

Memoniella Echinata.—Q.M.No. 10

Aspergillus Niger.—Q.M.No. 458

Aspergillus Clavatus.—Q.M.No. 862

4.4.5.1.3 *Procedure.*—Stock cultures to be used for the preparation of the spore-suspension inoculum shall be grown on the Potato-Dextrose Agar Medium, incubated at 85°F. and 90-95 percent relative humidity for 7 days, or until mature fruiting bodies are visible, and refrigerated until used. All culture not used after 21 days shall be discarded. The inoculum shall consist of a mixed spore suspension, aseptically prepared. Approximately 10 ml. of sterile distilled water shall be poured into each tube, the spores dispersed with the aid of a sterile needle, and the pure spore suspension from each tube added to 200 ml. of distilled water to which had been added 0.05 percent of a suitable wetting agent and previously sterilized. The morphological features of Chaetomium globosum preclude harvesting the spores in the above manner. The black hard fruit-bodies may be suspended in water and this suspension aseptically added to a milk dilution bottle containing 50 cc. of glass beads and 30 cc. of water which had been previously sterilized. Two or three minutes of vigorous shaking will release the spores from the hard perithecia and the resultant suspension can then be added to the flask containing the spore suspensions from each of the other five tubes.

Inoculation

The chemical constituents of the general test medium shall be carefully weighed and completely dissolved in the distilled water, after which the 2 percent agar is added, melted, and 15-ml. aliquots of the hot agar medium poured into petri dishes and the dishes sterilized at 120°C. and 15 pounds per

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square inch for 20 minutes. When the medium has hardened, the specimens shall be placed on the agar surfaces. The unsterilized specimens shall be placed in the sterile dishes as aseptically as possible. Raveled canvas strip specimens shall be folded approximately 1½ inches from each end, and the central portion placed on the agar surface. A minimum of five replicates shall be inoculated for each test exposure, and equal number shall be used as uninoculated controls.

The specimens shall be inoculated aseptically by pipetting 2 ml. of the mixed spore suspension onto the upper surface of that portion of the specimen lying on the surface of the agar. The inoculated specimens shall be incubated at 83° – 86°F. and a relative humidity of 90-95 percent. The duration of the incubation period shall be dependent upon the weight of the untreated material; material weighing 21 ounces or less per square yard shall be incubated 14 days; material weighing more than 21 ounces shall be incubated 21 days.

Evaluation

Quantitative.—At the end of the incubation period, the inoculated specimens shall be examined for visual growth of mildew. At the end of the incubation period, the inoculated specimens shall be removed from the petri dishes and gently rinsed in tap water to remove surface mildew. They shall be air-dried and conditioned for 24 hours at standard conditions of temperature and humidity. The uninoculated controls shall be similarly rinsed, air dried and conditioned. At the conclusion of the conditioning period, the percent of tensile strength retention shall be determined by averaging the tensile strength of the replicates in both the uninoculated and inoculated series for each test exposure. Breaking strengths shall be determined as described in method 5104 of Federal Specification CCC–T–191. Calculation of results shall be as described in method 5750 of Federal Specification CCC–T–191.

4.4.5.1.4 *Report.*—The report shall be as described in method 5750 of Federal Specification CCC–T–191, except that the results of the qualitative tests shall be "growth" or "no growth."

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Levels A, B, and C.*—The cotton duck cloth of one type and class only shall be packaged in accordance with the applicable requirements of Federal Specification PPP–P–51 for class 4 fabrics.

5.2 Packing.

5.2.1 *Levels A, B, and C.*—The cotton duck cloth of one type and class only shall be packed in accordance with the applicable requirement of Federal Specification PPP–P–51 for class 4 fabrics.

5.3 Marking.—In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of Federal Specification PPP–P–51.

6. NOTES

6.1 Intended use.

6.1.1 *Types I and II.*

6.1.1.1 *Class 1.*—Cotton duck treated for fire, water, weather, and mildew resistance as covered by this specification for class 1 is intended for use in manufacture of canvas covers, tents, tarpaulins, and other duck items providing protection under conditions of prolonged outdoor use.

6.1.1.2 *Class 2.*—Cotton duck treated for fire, water, weather, and mildew resistance as covered by this specification for class 2 is intended for use as upholstery material in motorized vehicles procured by Ordnance Department for the Department of the Army.

6.2 Ordering data. — Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

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(*a*) Title, number, and date of this specification.

(*b*) Base fabric required (see 3.4.1 and 6.8).

(*c*) Type and class required (see 1.2.1 and 3.5.5).

(*d*) Width of cloth required (see 3.5.2).

(*e*) Laboratory report, when not required (see 3.2).

(*f*) Flexibility requirements for duck other than numbered or Army ducks specified in 3.5.7.

(*g*) Mildew inhibitor (if other than specified in 3.5.10.1).

(*h*) Whether the cloth is to be packaged, packed, and marked for level A, level B, or level C (see 5.1).

6.3 Accelerated aging.—Where the accelerated aging apparatus specified in 4.4.3 is not available, other tests and requirements may be substituted provided data is submitted showing correlation with the accelerated weathering test of table VII.

6.4 Standard sample.—For access to standard sample, address the procuring office issuing the invitation for bids.

6.5 Mildew inhibitors.—The types of mildew inhibiting agents specified for type I fabrics were selected after lengthy and much more severe tests than those set forth in the specification. The specified test is only for the purpose of determining whether this treatment has been applied properly and is not to be considered as a performance test for the valuation of the effectiveness of the mildew inhibiting treatment. Consideration will be given to use of other mildew inhibiting agents or treatment which show good resistance to microbiological activity, a low degree of toxicity, and excellent retention of their efficiency following prolonged outdoor exposure, and are acceptable from a toxicological standpoint. At the time of submission of samples to the laboratory, a statement as to the kind of inhibitor used should be included in order to facilitate laboratory testing. Prior to procurement by the contracting agency, the chemical formulations of fire, water, weather, and mildew resistance compounds for all shall be reviewed from a skin toxicity standpoint and approval obtained from the office of the Surgeon General.

6.6 Gas composition. — A gas mixture known as Matheson Manufactured Gas B, which can be purchased from the Matheson Company, Inc., East Rutherford, New Jersey, conforms to requirements for the synthetic gas mixture specified in table VII.

6.7 Formula disapproval.—For the information of the contractor, the use of proteins and their derivatives, starch, hydrophylic materials, oxidizing oils and oxidizing resins and resin or compounds to increase hygroscopicity may lead to formula disapproval.

6.8 For the information of Procurement Officers it is pointed out that application of this finish affects a change in width from grey to finished state depending upon the width of material and the type of processing required. For guidance, table VIII may be used:

TABLE VIII.

Grey fabric width	Nominal decrease in width due to finishing	Material
34 inches maximum ----	½ inch	Plied yarn, Army and numbered Ducks
Over 34 inches--	1 inch	Plied yarn, Army and numbered Ducks
Up to 40 inches--	1½ inches	Flat Ducks requiring desizing
Over 40 inches--	2 inches	Flat Ducks requiring desizing

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6.9 This specification supersedes the following Military specifications:

MIL–D–10860(QMC), Duck, Cotton: Fire-, Water-, Weather-, and Mildew Resistant, 22 January 1952

MIL–C–17106A(Navy), Cloth, Cotton, Duck; Fire, Water, Mildew and Weather Resistant, 1 September 1953.

6.10 It is believed that Dark Green FWWMR Duck may be finished under the requirements for type I of this specification and that Duck (dark) Blue FWWMR Duck may be finished under the requirements for type II of this specification.

6.11 Transportation description. — Transportation description applicable to this item is:

Cloth, fire-proofed, oiled or waterproofed (not coated) not otherwise indexed by name:

Carload minimum weight — any quantity

Truckload minimum weight—30,000 pounds

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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OTHER INTEREST:
Army—M O Sig
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